* Maxwell's eqns, differential form, SI units

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\epsilon_0}, \quad \vec{\nabla}\cdot\vec{B} = 0, \quad \vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Maxwell didn't have $\vec{\nabla}$-notation. (8 equations)

$$\vec{\nabla}\times\vec{B} = \mu_0 \vec{J} + \mu_0\epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$\vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ / DISPL. CURRENT DENSITY

In Gaussian (CGS) units: $\vec{\nabla}\cdot\vec{E} = 4\pi\rho$, $\vec{\nabla}\cdot\vec{B} = 0$

$$\vec{\nabla}\times\vec{E} = -\frac{1}{c}\frac{\partial \vec{B}}{\partial t}, \quad \vec{\nabla}\times\vec{B} = \frac{1}{c}\frac{\partial \vec{E}}{\partial t} + \frac{4\pi}{c}\vec{J}$$

→ highlights symmetry betn $\vec{E}$ & $\vec{B}$

→ choice of units is a matter of tradition / convention / ~~conv~~ (in)convenience

→ $\mu_0\epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}$ comes from CONTINUITY EQ. (Maxwell's correction)

$\epsilon_0 \frac{\partial \vec{E}}{\partial t}$ is called DISPLACEMENT CURRENT ~~DENSITY~~

* Maxwell's eqns in integral form

$$\oint_\Sigma \vec{E}\cdot d\vec{S} = \frac{1}{\epsilon_0} Q_{encl.} = \frac{1}{\epsilon_0}\int_V \rho \, d^3r$$

[closed surface Σ encloses volume V]

$$\oint_\Sigma \vec{B}\cdot d\vec{S} = 0$$

$$\oint_C \vec{E}\cdot d\vec{l} = \int_{\Sigma'} \left(-\frac{\partial \vec{B}}{\partial t}\right)\cdot d\vec{S} = -\frac{d}{dt}\int_{\Sigma'} \vec{B}\cdot d\vec{S}$$

[closed path C encloses surface Σ']

G2

$$\oint_C \vec{B}\cdot d\vec{l} = \mu_0 I_{encl.} + \mu_0\epsilon_0 \int_{\Sigma'} \left(\frac{\partial \vec{E}}{\partial t}\right)\cdot d\vec{S}$$

$$= \mu_0 \int_{\Sigma'} \vec{J}\cdot d\vec{S} + \mu_0\epsilon_0\int_{\Sigma'}\left(\frac{\partial\vec{E}}{\partial t}\right) d\vec{S}$$

$$= \mu_0\int_{\Sigma'}\left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}\right)\cdot d\vec{S} = \mu_0 \int_{\Sigma'} \left(\vec{J} + \vec{J}_D\right)\cdot d\vec{S}$$

→ Deriving differential forms from integral forms: use Gauss's divergence theorem or Stoke's theorem

$$\oint_\Sigma \vec{V}\cdot d\vec{S} = \int_V \left(\vec{\nabla}\cdot\vec{V}\right) d^3r$$

$$\oint_C \vec{V}\cdot d\vec{l} = \int_{\Sigma'} \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{S}$$

* $\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t} = \mu_0\left(\vec{J}+\vec{J}_D\right)$

$\vec{J}_D = \epsilon_0\frac{\partial\vec{E}}{\partial t}$ is called the DISPLACEMENT CURRENT DENSITY (as opposed to $\vec{J}$ = conduction current density)

↑
Historical name, from Maxwell

* SCALAR & VECTOR POTENTIALS

We learned in electrostatics: $\vec{E} = -\frac{\partial V}{\partial t}$

because $\vec{\nabla}\times\vec{E} = 0$, [redacted]

In magnetostatics: $\vec{B} = \vec{\nabla}\times\vec{A}$ because $\vec{\nabla}\cdot\vec{B} = 0$

Now we are doing electrodynamics.

$\vec{\nabla}\cdot\vec{B} = 0$ is still true $\Rightarrow$ $\vec{B} = \vec{\nabla}\times\vec{A}$

But $\vec{\nabla}\times\vec{E} \neq 0$! Let's correct $\vec{E} = -\frac{\partial V}{\partial t}$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} = -\frac{\partial}{\partial t}\left(\vec{\nabla}\times\vec{A}\right) = -\vec{\nabla}\times\left(\frac{\partial\vec{A}}{\partial t}\right)$$

$$\Rightarrow \vec{\nabla}\times\left(\vec{E}+\frac{\partial\vec{A}}{\partial t}\right) = 0$$

$\Rightarrow$ We can define $\vec{E}+\frac{\partial\vec{A}}{\partial t} = -\vec{\nabla}V$

(64)

Thus
$$\boxed{\begin{aligned} \vec{E} &= -\vec{\nabla} V - \frac{\partial \vec{A}}{\partial t} \\ \vec{B} &= \vec{\nabla} \times \vec{A} \end{aligned}}$$

* GAUGE TRANSFORMATIONS

Certain Transformations ~~keep~~ of $(V, \vec{A})$ keep the physical fields $(\vec{E}, \vec{B})$ unchanged (invariant).

For any scalar field $\lambda(\vec{r}, t)$, the replacement

$$\boxed{\begin{aligned} \vec{A} &\longrightarrow \vec{A} + \vec{\nabla}\lambda \\ V &\longrightarrow V - \frac{\partial \lambda}{\partial t} \end{aligned}}$$

leave the fields $(\vec{E}, \vec{B})$ invariant

$\boxed{\text{Exercise : Show}}$

CULTURE : In the Special Theory of Relativity, Electricity & Magnetism are Unified even farther.

$(V, \vec{A})$ is combined to a single object (a 4-vector)

$(\rho, \vec{J})$ is " " " " 4-vector

$(\vec{E}, \vec{B})$ is combined to a single 4-Tensor.

* Maxwell's Eqs. in terms of Potentials:

2 & 3 automatically satisfied:

1: $$\nabla^2 V + \frac{\partial}{\partial t}(\vec{\nabla}\cdot\vec{A}) = -\frac{\rho}{\epsilon_0}$$

4: $$\left(\nabla^2\vec{A} - \mu_0\epsilon_0\frac{\partial^2\vec{A}}{\partial t^2}\right) - \vec{\nabla}\left(\vec{\nabla}\cdot\vec{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\vec{J}$$

Used $\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \nabla^2\vec{A}$

$\boxed{\text{Exercise : Show!}}$

* Coulomb gauge   Pick $\vec{\nabla}\cdot\vec{A} = 0$

Maxwell's Eqs. become:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad \left(\nabla^2\vec{A} - \mu_0\epsilon_0\frac{\partial^2\vec{A}}{\partial t^2}\right) - \mu_0\epsilon_0\vec{\nabla}\left(\frac{\partial V}{\partial t}\right) = -\mu_0\vec{J}$$

(Eq. for $\vec{A}$ not simple, if not in static situation.)

* Lorentz gauge $\vec{\nabla}\cdot\vec{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$

(same as Coulomb gauge in case of statics)

Maxwell's Eqs.: $$\left.\begin{aligned}\nabla^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} &= -\frac{\rho}{\epsilon_0}\\ \nabla^2\vec{A} - \mu_0\epsilon_0\frac{\partial^2\vec{A}}{\partial t^2} &= -\mu_0\vec{J}\end{aligned}\right\}\ \text{symmetric treatment}$$

$$\nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon_0} \quad , \quad \vec{\nabla}^2 \vec{A} - \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J}$$

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) = \Box^2 \quad ,$$ the d'Alembertian operator

$$\Box^2 V = -\frac{\rho}{\epsilon_0} \quad , \quad \Box^2 \vec{A} = -\mu_0 \vec{J}$$

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* The Displacement Current

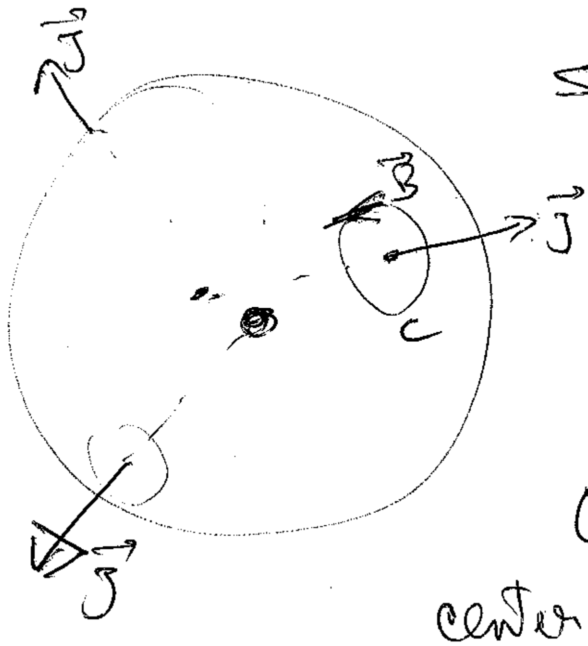
(Maxwell's new term)

$$\vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

corrects $\vec{\nabla}\times\vec{B} = \mu_0 \vec{J}$ (Ampère's law)

$$\rightarrow \vec{\nabla}\times\vec{B} = \mu_0(\vec{J}+\vec{J}_D)$$

Example 1



Imagine charge being emitted, spherically symmetrically, from a source.

(Could be a radioactive source)

Consider distance R from center of source, imagine sphere.

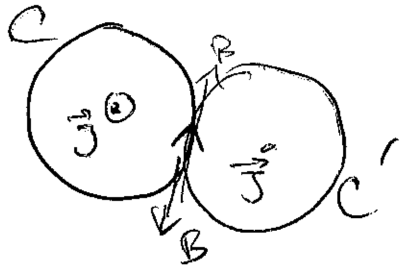
Using Ampere's law, $\oint_C \vec{B}\cdot d\vec{l} = \mu_0 I_{enc.}$,

At this distance,
$J(R) = \frac{1}{4\pi R^2}\frac{dQ}{dt}$

we predict $\vec{B}$-field along C.

$\Rightarrow$ But this breaks symmetry. E.g., choosing C', you find $\vec{B}$ in opposite direction!

$\Rightarrow$ $\vec{B}$ must be zero.



Q. Current does not produce $\vec{B}$?
$\vec{\nabla}\times\vec{B} = \mu_0 \vec{J}$ violated?

A. In this case, $\vec{J}$ is balanced by $\vec{J}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$. The charge inside sphere creates field $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^2}$ (Gauss's law)

$$\Rightarrow J_D = \epsilon_0 \frac{\partial E}{\partial t} = \frac{\epsilon_0}{4\pi\epsilon_0 R^2}\frac{dQ}{dt} = \frac{1}{4\pi R^2}\left(-4\pi R^2 J\right)$$

$$= -J$$

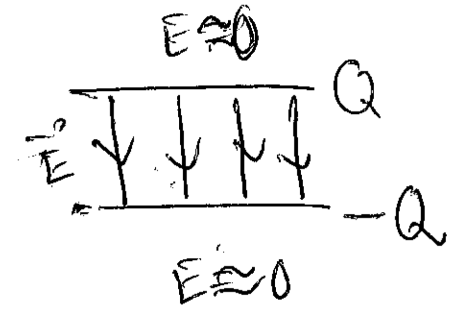
so that $\vec{\nabla}\times\vec{B} = \mu_0\left(\vec{J} + \vec{J}_D\right) = 0$

optional

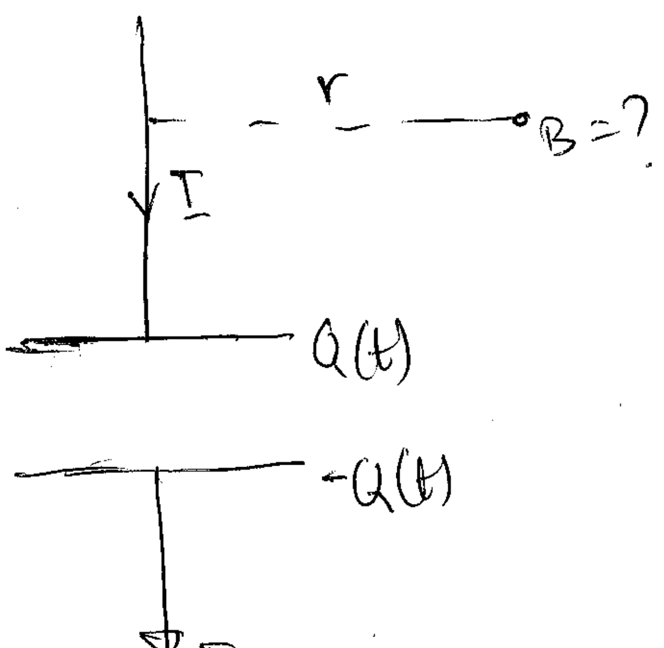
Example 2

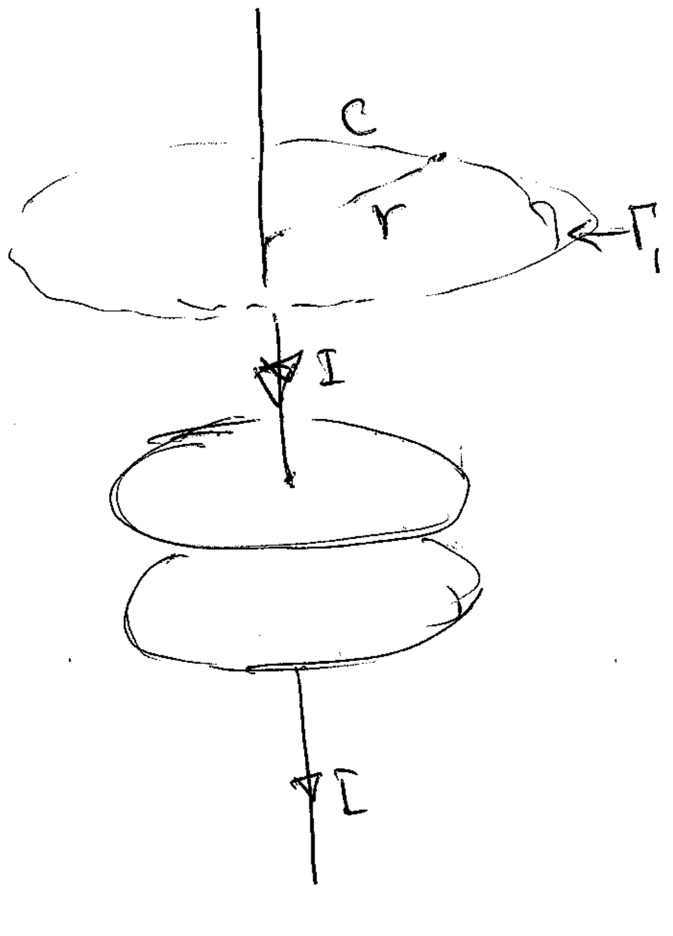
Charging a parallel-plate capacitor (a.k.a. condenser)

Look up Feynman chapter 18



capacitor

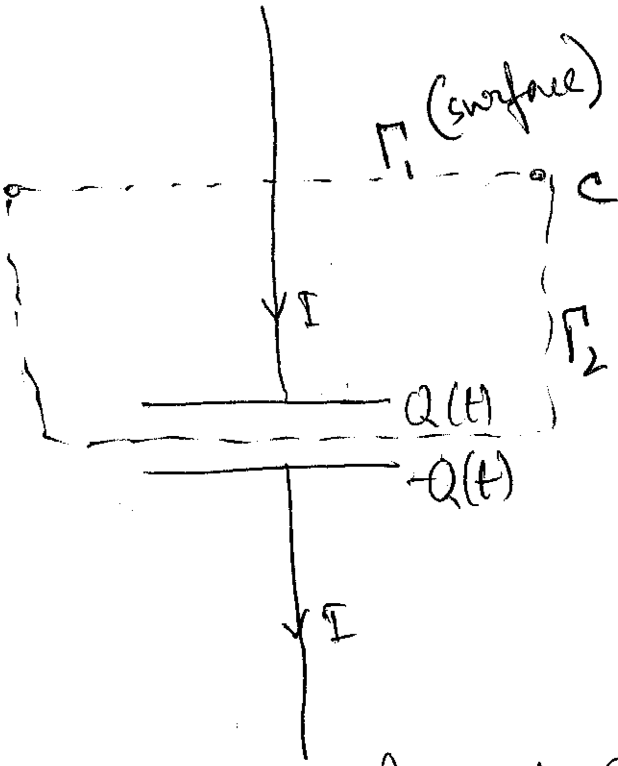




Ampere's law gives

$$\oint_C \vec{B}\cdot d\vec{l} = B \cdot 2\pi r = \mu_0 I \quad \Rightarrow \quad B = \frac{\mu_0 I}{2\pi r}$$

$I =$ ~~net~~ current thru Γ_1



Surface Γ_2 also has C as boundary. No current through Γ_2 $\Rightarrow$ Ampere's law gives zero field at same point!

Ampere: $\oint_C \vec{B}\cdot\vec{dl} = B\cdot 2\pi r = \int_{\Gamma_2} \vec{J}\cdot\vec{dS} \cancel{= 0}$

$= I_{2,\text{encl.}} = 0$

Solution: Maxwell's term. If considering Γ_2 the displacement ~~term~~ current generates B-field.

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 \int_{\Gamma_2} \vec{J}\cdot\vec{dS} + \mu_0\epsilon_0 \int_{\Gamma_i} \frac{\partial \vec{E}}{\partial t}\cdot\vec{dS}$$

($\Gamma_i = \Gamma_1$ or Γ_2)

If surface is Γ_1,

$$\oint \vec{B}\cdot\vec{dl} = \mu_0 I + 0 \quad \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

If surface is Γ_2,

$$\oint \vec{B}\cdot d\vec{l} = 0 + \mu_0\epsilon_0 \frac{d}{dt}\int_{\Gamma_2} \vec{E}\cdot d\vec{S}$$

$$= \mu_0\epsilon_0 \frac{d}{dt}\left(\frac{Q}{\epsilon_0}\right) \quad \text{Gauss's law}$$

$$= \mu_0 \frac{\partial Q}{dt} = \mu_0 I$$

$\Rightarrow$ Same answer, $B = \frac{\mu_0 I}{2\pi r}$,
whichever surface is used.

★ Maxwell's Eqs in Free Space/Vacuum

$\longrightarrow$ Electromagnetic waves

$\boxed{\rho = 0,\ \vec{J} = 0 \ \text{no charge or current}}$ (wave solution of Max. Eqs.)

$$\vec{\nabla}\cdot\vec{E} = 0, \quad \vec{\nabla}\cdot\vec{B} = 0, \quad \vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{dt},$$

$$\vec{\nabla}\times\vec{B} = \mu_0\epsilon_0 \frac{\partial \vec{E}}{dt} = \frac{1}{c^2}\frac{\partial \vec{E}}{dt}$$

The curl equations, indicate, $\vec{E}$ & $\vec{B}$ fields could "keep going" without charges or currents, once initiated.

More symmetric:

$$\vec{\nabla}\times\vec{E} = -\frac{1}{c}\frac{\partial(c\vec{B})}{dt}$$

$$\vec{\nabla}\times(c\vec{B}) = \frac{1}{c}\frac{\partial \vec{E}}{\partial t}$$

in Gaussian units $c\vec{B}$ is called $\vec{B}$.

* WAVE EQUATIONS FROM MAXWELL'S EQUATIONS

Reminder: — $\frac{\partial^2 f}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} = 0$

is a "wave equation". Any function of the form $f(x,t) = \phi(x - vt)$ or $\phi(x+vt)$ is a solution. are traveling wave solutions. E.g., → traveling with speed v.

$$f(x,t) = f_0 \sin(x \pm vt), \quad f(x,t) = f_0 \cos(x \pm vt)$$

are solutions of the wave equation.

f could be: displacement of string, water surface, etc.

* Now, wave equations for $\vec{E}$ & $\vec{B}$:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t}\right)$$

$$\Rightarrow \underbrace{\vec{\nabla}(\vec{\nabla}\cdot\vec{E})}_{0} - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\underbrace{(\vec{\nabla} \times \vec{B})}_{+\frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}}$$

$$\Rightarrow \boxed{\nabla^2 \vec{E} = \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2}}$$

This is a 3D version of the wave equation:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

(propagation can be in any direction)

The object propagating is a vector.

$\vec{B}$-field obeys same equation in free space:

$$\vec{\nabla}\times\vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} \Rightarrow \vec{\nabla}\times(\vec{\nabla}\times\vec{B}) = \vec{\nabla}\times\left(\frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}\right)$$

$$\Rightarrow \vec{\nabla}\underbrace{(\vec{\nabla}\cdot\vec{B})}_{=\,0} - \nabla^2\vec{B} = \frac{1}{c^2}\frac{\partial}{\partial t}\underbrace{(\vec{\nabla}\times\vec{E})}_{=\,-\frac{\partial \vec{B}}{\partial t}}$$

$$\Rightarrow \boxed{\nabla^2\vec{B} = \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2}}$$

* Thus $\vec{E}$ and $\vec{B}$ in free space satisfy

$$\nabla^2\vec{E} - \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} = 0, \quad \nabla^2\vec{B} - \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

* In Lorentz gauge, potentials in free space ($\rho = 0$, $\vec{J} = 0$) also satisfy

$$\nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = 0, \quad \nabla^2\vec{A} - \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = 0$$